# Business Requirements Document (BRD)

## Project: Premium Food Shop E-Commerce Platform

### 1. Executive Summary & Business Nature

The proposed system is a comprehensive, end-to-end e-commerce web platform designed for a physical food retail business. The business aims to expand its reach by offering a digital storefront where customers can explore the product catalog, view special promotions, and place orders online seamlessly.

The primary objectives of this digital transformation are:

- **Enhanced Customer Experience:** Providing a user-friendly, aesthetically pleasing platform to browse high-quality food products and seamlessly place orders.
- **Operational Automation:** Streamlining the order management process from the moment a customer places an order to its fulfillment, reducing manual administrative tasks.
- **Brand Trust & Communication:** Utilizing automated PDF invoicing and real-time WhatsApp notifications to keep customers informed and build a professional brand image.

### 2. User Roles

The system will cater to two primary user entities:

1. **Customer (End-User):** Can browse the site as a guest or a registered user to view products, apply offers, and place orders.
2. **Store Administrator (Admin):** The business owner or manager who operates the backend portal to manage inventory, process orders, and oversee business analytics.

### 3. Functional Requirements

#### 3.1. Customer Portal (Front-End)

The front-end must focus on a modern, mobile-first UI/UX design, ensuring high conversion rates and ease of use.

- **Dynamic Homepage:**
  - Hero banner showcasing ongoing promotions and special offers.
  - Sections for "Best Sellers," "New Arrivals," and "Featured Categories."
- **Product Catalog & Advanced Search:**

    - Categorization of food items (e.g., Snacks, Sweets, Beverages, Bakery).
    - Search bar with auto-suggest functionality.
    - Filtering options (by price, category, dietary preference like Veg/Non-Veg).
- **Product Detail Page (PDP):**
    - High-resolution image gallery with zoom functionality.
    - Comprehensive details: Price, weight/portion size, detailed description, ingredients, and allergy warnings.
    - Availability status (In Stock / Out of Stock).
- **Promotions & Offers:**
    - Dedicated "Special Offers" page.
    - Ability to apply Promo Codes during checkout.
    - Visual indicators (e.g., strike-through prices, discount badges) on discounted items.
- **Shopping Cart & Checkout:**
    - Fly-out or persistent cart showing added items, subtotal, and delivery charges.
    - Multi-step checkout process (Contact Info -> Delivery Details -> Order Summary).
- **Order Confirmation & PDF Generation:**
    - Upon successful order placement, a dynamic PDF receipt must be generated instantly.
    - The PDF must include: Order ID, Date/Time, Itemized list with quantities and prices, Total Amount, Customer details, and Shop contact information.
    - A prominent "Download Invoice" button on the "Thank You" page.

### 3.2. Admin Portal (Back-End Dashboard)

A secure, authenticated portal for business management with a premium, data-rich UI.

- **Secure Authentication:**
    - Admin login page with encrypted credentials and session management.
    - Password recovery mechanism.
- **Super Dashboard (Analytics & Insights):**
    - Visual summary of key metrics: Total Orders (Today/Month), Total Revenue, Pending Orders.
    - Graphical charts showing sales trends.
    - List of top-selling products and low-stock alerts.
- **Product Management:**

    - CRUD operations (Create, Read, Update, Delete) for all food items.
    - Ability to upload and manage product images.
    - Inventory management: Updating stock counts and toggling visibility.
- **Offer & Promotion Management:**
    - Create site-wide banners for sales.
    - Generate and manage promo codes (discount percentage, fixed amount, expiry dates).
- **Order Management System (OMS):**
    - Real-time datatable showing all orders with statuses (Pending, Accepted, Dispatched, Delivered, Canceled).
    - Detailed view of each order (Customer info, items ordered, total bill).
    - Action buttons to change order status (e.g., "Accept Order").
- **WhatsApp API Integration:**
    - Automated trigger: When the Admin clicks "Accept Order," the system pushes a webhook to the WhatsApp Business API.
    - The customer receives an automated WhatsApp message: *"Hello [Name], your order # [OrderID] has been accepted by [Shop Name] and is being processed. Total: [Amount]. Thank you!"*

**4. Non-Functional Requirements (Technical & UX)**

- **UI/UX Standards:** * Clean, minimalist design utilizing high-quality whitespace, appetizing color palettes (e.g., warm reds, oranges, or fresh greens), and modern typography.
    - 100% Mobile Responsive design (as over 70% of e-commerce traffic is mobile).
- **Performance:**
    - Fast page load times (under 3 seconds) leveraging image optimization and caching.
- **Security:**
    - SSL Certificate (HTTPS) encryption for all data transfers.
    - Protection against SQL injection and Cross-Site Scripting (XSS).
- **Scalability:**
    - The database and hosting architecture should support sudden spikes in traffic (e.g., during festive seasons or major discount campaigns).

**5. Future Enhancements (Phase 2)**

*(These are features that can be added later once the core system is stable)*

- Integration with local Payment Gateways (e.g., PayHere, Webxpay) for direct online card payments.
- Live Delivery Tracking via GPS.
- Customer Loyalty Points System to encourage repeat purchases.
- Multi-language support (English, Sinhala, Tamil) based on customer preference.